Identification of molecules by structure

We were given a sample of complex organic molecules having an equal count of atoms and bonds. Furthermore, each molecule in the sample fulfils the following criteria:

- Every atom is bonded to one, two, or three other atoms.
- Some atoms form cycles.
- If an atom has exactly two bonds, it is always a member of a cycle.
- Every cycle comprises exactly one atom with three bonds.

Our objective is to identify the number of distinct structures present within the sample. Two molecules exhibit the same structure if a one-to-one mapping exists among their atoms while preserving bonds. In such cases, we say that these molecules are *structurally identical*. (Note that this definition does not consider atomic elements or the types of their bonds.)

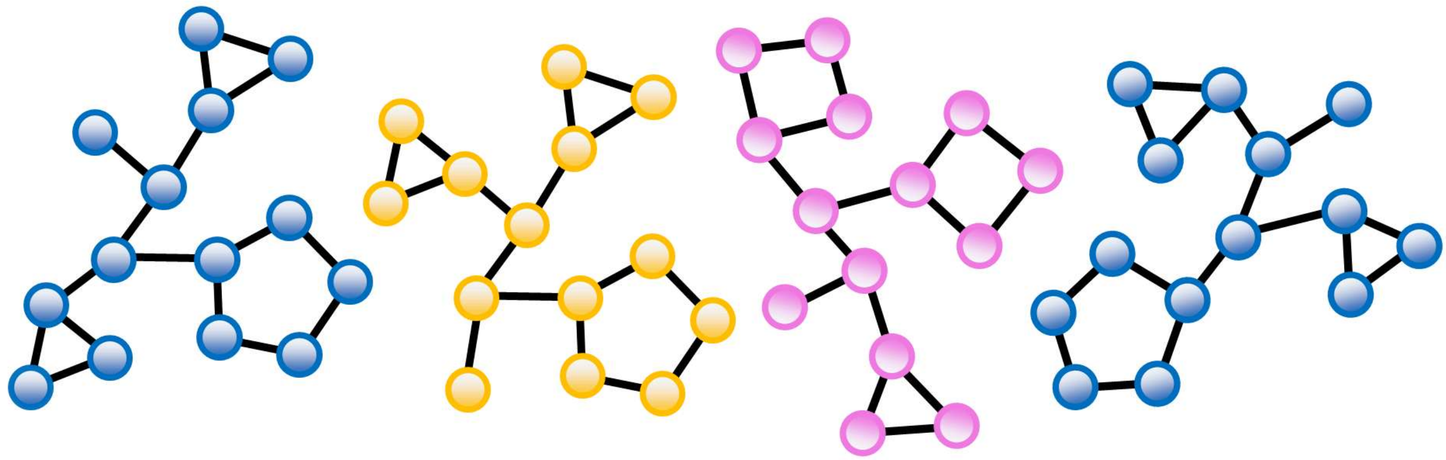

Figure 1. Illustration showing four molecules, each composed of 14 atoms (depicted as nodes) and 16 bonds (depicted as edges). Every molecule has 3 cycles within its structure. The only pair of structurally identical molecules are the first and last molecules drawn in blue.

The task

For a given sample represented by a set of undirected graphs, sort the molecules into groups by their structure and print out the counts of structurally identical molecules for each detected class.

Input

The input line contains three integers M, A, and B, separated by spaces. Here, M represents the number of molecules. Each molecule comprises A atoms and B bonds. Following this, there are M lists, where each list specifies B bonds for an individual molecule. Each list contains B consecutive lines, with each line comprising two integers A_1 and A_2, separated by a space. This pair denotes a bond between atoms A_1 and A_2. For each molecule, its atoms are arbitrarily numbered from 1 to A.
It holds $1 \leq M \leq 1000$, $4 \leq A \leq 9000$ and $A \leq B \leq 10000$.

Output

The output is one line containing N integers C_1, ..., C_N separated by spaces. These integers are arranged in the ascending order. N is the total number of structural classes detected among the molecules. Each C_i is the number of structurally identical molecules belonging to the i-th class. Hence, the sum of C_i's is M.

Example 1

Input

```
2 9 10
7 6
7 8
8 6
6 5
4 5
3 4
4 1
3 2
1 2
5 9
8 7
6 8
7 6
3 5
5 6
3 1
3 4
2 1
2 4
9 5
```

Output

```
2
```

Example 2

Input

```
4 14 16
13 8
7 8
7 13
1 7
2 1
11 5
5 2
5 6
6 9
9 10
10 11
3 14
12 1
4 14
4 2
4 3
2 6
5 1
3 5
6 10
10 1
2 5
4 3
11 3
13 14
13 12
12 11
12 14
7 9
8 9
8 11
8 7
6 5
6 7
7 8
8 9
2 9
8 5
11 12
11 10
10 9
4 10
4 12
13 14
13 3
2 1
2 3
3 14
4 11
12 11
3 13
3 12
13 12
5 9
9 6
10 11
7 6
5 10
8 7
8 5
10 14
1 14
1 2
2 14
```

Output

```
1 1 2
```

The data of Example 2 is depicted in Figure 1.

Public data

The public data set is intended for easier debugging and approximate program correctness checking. The public data set is stored also in the upload system and each time a student submits a solution it is run on the public dataset and the program output to stdout and stderr is available to him/her.
[Link to public data set]